

Behind the Screens: A Replication and Extension of Coasian Bargaining Experiments in the Digital Age^{*}

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Abstract

This paper replicates Hoffman and Spitzer's seminal Coasian bargaining experiments from the early 1980s and extends them to examine the impact of digital communication. We find that, while the face-to-face replication results mostly align with the original findings, transitioning to a digital environment induces a 23.3 percent decrease in efficient decision-making and over a fourfold increase in self-regarding behavior. These effects are amplified in one-shot bargaining scenarios and when property rights are strengthened and persist as bargainers gain experience. Our findings allude to several implications of digital communication for efficiency and welfare distributions in negotiation settings.

JEL Classifications: D9, D23, C78, J52, D83, D61, D62, K00, Q00

Key Words: Coase, bargaining, experiment, replication, online, communication, efficiency, sharing

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1 Introduction

Coase (1960) transformed the externality problem by proposing that well-defined property rights could lead to allocative efficiency, assuming no transaction costs or information asymmetries. Hoffman and Spitzer (HS hereafter) (1982; 1985) conducted the first experimental tests of Coase's theory. This paper builds on HS's work with two main goals: replicating their seminal experiments and exploring the robustness of bargaining behavior when transitioning from a face-to-face to digital environment.

HS's experiments strongly supported Coase's proposal, demonstrating that bargaining under Coasian conditions nearly always led to Pareto-efficient allocations. This held true regardless of how property rights were assigned (randomly or through competition), whether subjects bargained once or twice, and whether entitlement language was used to reinforce property rights. Additionally, HS found that property rights holders (Controllers) were more other-regarding in two-shot bargaining settings with randomly assigned property rights, but more self-interested in one-shot bargaining or when property rights were strengthened through competition and entitlement priming. Hoffman et al. (1994; 1996) suggested that the lack of anonymity in face-to-face settings likely contributed to these cooperative outcomes.

Our replication of HS's face-to-face treatments yields four main results. First, we fully replicate HS's efficiency results, finding equally high efficiency in all treatments. Second, we replicate the finding that Controllers in two-shot bargaining are other-regarding, though we do not find a significant difference between one- and two-shot bargaining. Third, we replicate HS (1985) in finding high efficiency in two-shot bargaining with both strong and weak property rights treatments, though we observe more sharing under strong property rights than HS. Fourth, using HS's greed index, we replicate the finding that strong property rights induce more self-regarding behavior than weak property rights.

As digital communication has become increasingly prevalent, it raises questions about how changes in communication media impact negotiation behavior and outcomes. While digital communication offers speed, convenience, and accessibility, it also introduces challenges like reduced social cues and increased social distance, which can harm communication quality, trust, and cooperation (Keisler et al. 1984; Rice

1987; Hiltz et al. 1986; Frohlich and Oppenheimer 1998; Brosig et al. 2003; Naquin and Paulson 2003; Bicchieri and Lev-On 2007; Charness and Gneezy 2008; Conrads and Lotz 2015; Cohn et al. 2022).^{1,2} Still, studies show mixed results. For instance, Croson (1999) found no efficiency loss, Greiner et al. (2014) found higher cooperation (and potentially efficiency gains) in face-to-face than digital settings, and Abatayo et al. (2018) demonstrated that young adults perform equally well in online and in-person settings. Similarly, Galeotti et al. (2019) found that participants prioritize efficiency over equality in online bargaining, while Bochet et al. (2006) observed high cooperation in both digital and face-to-face contexts when communication was meaningful. Nevertheless, face-to-face settings appear more effective for complex exchanges and fairness.

Our extension of HS's experiments to a digital bargaining environment yields four main results. First, digital bargaining reduces efficiency by 23.3 percent. Second, Controllers become over four times more self-regarding, as measured by the greed index. Third, this drop in efficiency is driven by increased self-regarding behavior, with digital Controllers consistently denying proposals from Bargainers (subjects without property rights) that would increase efficiency but require them to sacrifice earnings or bargain to achieve a payoff at least as great as their unilateral maximum. Fourth, while efficiency improves in both environments as subjects gain experience, a notable gap remains: nearly 100 percent of bargains reach Pareto efficiency in later face-to-face rounds, compared to about 80 percent for digital treatments.

These findings suggest a tradeoff exists between efficiency and individual rationality. In face-to-face settings, Controllers forgo earnings to maintain cooperation, whereas digital bargaining restores individual rationality at the cost of overall efficiency. This pattern persists across treatments, except in cases of repeated interactions with weak property rights.

¹ In contrast, face-to-face interactions enhance cooperation by reducing social distance and fostering other-regarding behavior (Ostrom and Walker 1991; Hoffman et al. 1996; Bohnet and Frey 1999; Brosig et al. 2003; Conrads and Reggiani 2016).

² Cooper and Kael (2009) and Drouvelis (2021) provide comprehensive summaries of the literature on social preferences in experiments and other decision settings. Bordia (1997) reviews early experimental studies comparing face-to-face and computer-mediated communication, while Geiger (2020) offers a theoretical overview and six decades of empirical research on communication media and negotiation.

2 Replication

2.1 Detailing the Original Studies: Hoffman and Spitzer (1982; 1985)

This section describes the experimental Coasian bargaining protocols implemented in HS (1982; 1985), since they are the basis for this study. HS (1982) featured groups of two subjects who were randomly assigned to bargaining role A or B.³ Subjects within a group negotiated under full and symmetric information over seven possible allocations listed in a payoff table (Table 1).⁴ The experimenters randomly assigned property rights to either player A or B (called the “Controller”), giving this subject the authority to end negotiations and implement their preferred allocation unilaterally.

Table 1. Sample Payoffs from HS (1982).

Decision Table		
<i>Number</i>	<i>Payoff to Player A</i>	<i>Payoff to Player B</i>
0	0.00	12.00
1	4.00	10.00
2	6.00	6.00
3	8.00	4.00
4	9.00	2.00
5	10.00	1.00
6	11.00	0.00

The remaining subject (called the “Bargainer”) could “attempt to influence the Controller to reach a mutually acceptable joint decision; offer to pay part or all of their earnings to the Controller” (HS 1982, p. 83). Notice that the Controller can unilaterally choose an equal split (allocation 2) or unilaterally maximize their own payoff (allocation 0 for B or 6 for A). The experiment tests whether A and B can agree

³ HS also conducted treatments with three-person groups that involved one Controller and two non-Controllers but were otherwise the same. We did not replicate the three-person games, and do not discuss these treatments further.

⁴ This experiment used information as an experimental factor. Players had either full or asymmetric information about payoffs, where each player only knows their own payoffs under the latter. We do not replicate this information factor.

to a transfer that instead facilitates the Pareto efficient allocation (number 1). If so, both parties signed a written document stating the agreement. HS (1982) explored how both repeated bargaining and information structure influenced bargaining efficiency and resulting allocations. However, we adopt only the former, where bargaining pairs engaged in repeated (two-shot) or one-shot bargaining.⁵

Results from HS (1982) strongly support the efficiency prediction of Coase (1960), with 95 percent of pairs choosing the joint-payoff-maximizing number. Repeated interaction also led to more frequent equal divisions (90 percent of repeated bargains vs. 33 percent of one-shot bargains).⁶ Observing that Controllers often sacrificed earnings for more equal divisions, HS (1985) further investigated the roles of entitlement and fairness in determining payoff distributions. They hypothesized that randomly assigned property rights failed to create a moral basis for self-regarding behavior. This second study used identical experimental protocols but introduced two methods of reinforcing property rights: subjects either earned property rights through competition or received them randomly, and instructions used neutral or entitlement language when discussing property rights. The results from HS (1985) are provided in Table 2.

Table 2. Efficiency and Sharing Outcomes from HS (1985).

	Neutral Language		Entitlement Language	
Random Entitlement	<i>N</i>	22	<i>N</i>	20
	<i>Efficient</i>	20 (.91)	<i>Efficient</i>	19 (.95)
	<i>Equal Division</i>	10 (.45)	<i>Equal Division</i>	9 (.47)
Game Entitlement	<i>N</i>	22	<i>N</i>	22
	<i>Efficient</i>	18 (.82)	<i>Efficient</i>	21 (.95)
	<i>Equal Division</i>	9 (.41)	<i>Equal Division</i>	4 (.18)

Notes: The leftmost column indicates the property rights assignment mechanism used. The top row of each property rights assignment indicates the total number of decisions made in each property rights-language treatment cell. Below the number of decisions made in each treatment cell, we present the total number of efficient decisions made, the total number of decisions where an equal division of the available surplus was realized, and in parentheses the fraction of the total number of decisions that these outcomes constitute.

⁵ The second factor in HS (1982) is information: both players have either full or asymmetric information about payoffs, where each player only knows their own payoffs under the latter.

⁶ Most bargaining outcomes in the repeated interaction setting involved an equal splitting of money.

Similar to HS (1982), HS (1985) reported high efficiency rates, with 91 percent of pairs (78 of 86) selecting the payoff-maximizing number. Efficiency also remained robust across treatments, unaffected by competition for property rights or entitlement priming. However, combining competition with entitlement language significantly reduced equal divisions. Their measure of payoff inequality, the “Average Greed Index,” indicated that each of these treatments led to more self-regarding behavior among Controllers.

2.2 Experimental Design

We implemented the full-information, two-person face-to-face bargaining protocols introduced by HS (1982; 1985) in a 2x2 between-subjects replication experiment, varying two factors: repeated bargaining (one-shot or two-shot, as in HS 1982) and the strength of property rights (weak or strong, as in HS 1985).⁷ To extend this design, we introduced a third factor—digital bargaining—enabling us to study the effects of changing the communication medium and gain a richer understanding of how the digital environment interacts with structural features that influence bargaining outcomes.

In contrast to HS (1982; 1985), subjects in our experiment made 10 bargaining decisions, allowing us to study how experience affects allocative efficiency and welfare distributions in both environments. In one-shot sessions, subjects bargained with 10 different partners, whereas in two-shot sessions, they bargained twice with 5 different partners. Although the payoff tables changed after each bargaining decision, the overall structure of payoffs remained the same as in Table 1.⁸

We recruited undergraduate students from Texas A&M University using ORSEE (Greiner 2004) and conducted 16 sessions (2 per treatment) with 12 subjects per session (24 per treatment) between December 2016 and September 2017. All treatments were randomized at the session level. For each session, we randomly selected two bargaining decisions for payment. In two-shot sessions, subjects were paid for

⁷ Note that instead of replicating all four treatments from HS (1985), we focus on the two most extreme treatments: strong property rights (competing for rights and entitlement priming) and weak property rights (randomly assigning rights and no entitlement priming).

⁸ See Appendix C for a list of all 10 payoff tables used.

both decisions made with a single partner, while in one-shot sessions, subjects were paid for two decisions made with two different partners.⁹

Procedures

We employed complete-stranger matching for both one- and two-shot bargaining sessions to ensure that a bargaining pair never matched more than once in a session (see Appendix A.1 for a full description of face-to-face matching protocols). The lab was arranged to maximize privacy, with spaced-out experiment stations and dividers.¹⁰ The same moderator read instructions aloud for all sessions, with written instructions provided for reference.¹¹ Instructions concluded with a comprehension quiz, which was checked individually before proceeding. During bargaining, Controllers always had the ability to unilaterally choose an allocation (i.e., to make decisions absent trade and without input from their counterpart) or to negotiate offers from the Bargainer and transfer money between one another. Once a pair finished bargaining and completed the contract in a period, they signaled an experimenter to collect the payoff tables and contract.¹² Subjects then waited quietly until all pairs finished bargaining before proceeding to the next period. After completing all bargaining periods, subjects completed a brief survey before exiting the lab.

Assigning Property Rights

We allocated *weak property rights* randomly via coin flip at the pair level. If the result was heads, the subject with the lower identification number in each pair was told they were *designated* as the Controller for that period (we assigned each subject a unique identification number between 1 and 12 during check-in). We allocated *strong property rights* by having subject pairs play a deterministic hash mark game (see Appendix A.2), and the winner was told they had *earned the right* to be the Controller for that period.¹³

⁹ In the nine sessions that took place between December 2016 and May 2017, we paid subjects a \$5 show-up fee. In the seven sessions conducted in September 2017, we paid subjects a \$10 show-up fee due to a change in lab policy.

¹⁰ About 15 feet of distance separated each station, helping to prevent bargaining parties from overhearing each other's bargaining strategies and ensuring privacy from experimenter scrutiny.

¹¹ See Appendices A.2 and A.3 for the instructions used in all face-to-face sessions.

¹² See Appendix A.4 for the agreement form that bargaining pairs filled out and signed after finishing each decision.

¹³ We asked subjects to record a strategy for this game. There is no evidence that any subject solved the game.

2.3 Results: Face-to-Face Replication

We consider a finding replicated if we obtain a significant result in the same direction as HS, which is the most rigorous standard used in Camerer et al. (2016). Following HS, *sharing* is defined as any allocation where Controller and Bargainer payoffs are within \$1 of equality. We use Fisher's exact tests to assess statistical differences in the proportions of efficient and sharing allocations, both within and across studies.¹⁴ Table 3 presents the results of our replication of HS (1982; 1985), using data from only the first two decisions made by each subject pair in our face-to-face sessions.¹⁵ The set of replication results for the two-person, full-information treatments in HS (1982) are as follows.

Result 1: We fully replicate HS's efficiency results. Fisher's exact tests indicate no significant differences in the proportions of efficiency achieved between the two studies in both in one-shot ($p = 0.12$) and two-shot ($p = 0.11$) bargaining.

Result 2: We replicate the finding that Controllers in two-shot bargaining are other-regarding, with no significant difference in the proportion of sharing allocations between the two studies ($p = 1.00$). However, unlike HS, we do not find that repeated bargaining increases other-regarding behavior, as there is no significant difference between sharing decisions in our one- and two-shot treatments ($p = 0.36$).¹⁶

Next, we focus on replicating HS (1985), also in Table 3, which studies the effect of property right strength on bargaining behavior.

Result 3: We replicate the finding that the strength of property rights does not affect efficiency, with equally high levels of efficiency observed in both strong and weak property rights treatments ($p =$

¹⁴ Fisher's exact test is a proportions test that is designed for use within small samples. HS find no statistically significant difference in the number of efficient decisions in their one- and -two shot bargaining environments, but they do find a difference in the number of sharing decisions.

¹⁵ Power calculations indicate that a sample size of 16 observations per treatment is sufficient to detect an effect similar to HS (1982)'s result for sharing allocations between the one-shot and two-shot treatments, while a sample size of 34 observations per treatment is needed to detect an effect similar to HS (1985)'s result for sharing allocations between the strong and weak property rights treatments, both to achieve 80% power at a 5% significance level. Consequently, we are underpowered for the latter replication without including more periods or pooling with one-shot treatments.

¹⁶ Note that when using an equal split definition of sharing, our results do not change across one-shot and two-shot environments. However, the statistical difference in the proportion of sharing between one- and -two shot bargaining in HS disappears as there are four fewer sharing decisions in their two-shot bargaining sessions under this definition.

0.34). However, we fail to replicate the finding in HS (1985) that stronger property rights reduce the proportion of sharing allocations ($p = 0.14$). Although we observe a higher proportion of sharing allocations under strong property rights than HS, this difference is not statistically significant ($p = 0.245$).

Table 3. Comparing the Impacts of Repeated Bargaining and the Strength of Property Rights on Efficiency and Sharing Allocations in Our Replication to those HS (1982; 1985).

			<i>Efficient Decisions</i>			<i>Sharing Decisions</i>		
Bargaining Sequence	Property Rights	$N_{HS} N_{Our}$	HS Data	Our Data	Fisher's Exact	HS Data	Our Data	Fisher's Exact
Replicating HS 1982								
1-Shot	Weak	12 24	11 (.92)	15 (.63)	$p = 0.12$	5 (.42)	14 (.58)	$p = 0.48$
2-Shot	Weak	34 24	32 (.94)	19 (.79)	$p = 0.11$	26 (.76)	18 (.75)	$p = 1.00$
Fisher's Exact	1-Shot vs. 2-Shot		$p = 1.00$	$p = 0.34$		$p = 0.04$	$p = 0.36$	
Replicating HS 1985								
2-Shot Bargaining	Strong	22 24	21 (.95)	15 (.63)	$p = 0.011$	7 (.32)	12 (.50)	$p = 0.245$
2-Shot Bargaining	Weak	22 24	20 (.91)	19 (.79)	$p = 0.418$	14 (.64)	18 (.75)	$p = 0.525$
Fisher's Exact	Strong vs. Weak		$p = 1.00$	$p = 0.34$		$p = 0.07$	$p = 0.14$	

Notes: This table presents the number of Pareto efficient and sharing decisions (proportion of total decisions in parentheses) by session type in both our and HS's experiments using the first two periods of bargaining in all cells, with p -values for Fisher's exact tests. In the HS (1982) replication panel, the one-shot bargaining cells include 12 observations from HS (1982), and the two-shot bargaining cells include 12 observations from HS (1982) and 22 from HS (1985). Our results for cells in this panel include data from our face-to-face, coin flip (weak property rights), no entitlement priming sessions ($N = 24$ for each one-shot and two-shot cell). In the HS (1985) replication panel, all HS results in the two-shot, strong and weak property rights cells include 22 observations from HS (1985). Our results for cells in this panel include data from our face-to-face, two-shot, strong and weak property rights sessions ($N = 24$ for each strong and weak property rights cell).

HS (1985) introduced the Average Greed Index (AGI) to further measure the effect of property right strength on self-regarding behavior. The AGI represents the average difference between a Controller's final payoff and an equal split of the final payoff.¹⁷ In Table 4, we use data from the first two periods of our two-shot, weak- and strong-property rights sessions to compare to HS (1985).

Result 4: As measured by the AGI, we replicate the finding that strong property rights induce more self-regarding behavior. In HS (1985), strong property rights produced an AGI of \$4.52 and weak property

¹⁷ Hence, an $AGI > 0$ indicates a payoff favoring the Controller, $AGI = 0$ indicates an equal split, and $AGI < 0$ indicates a payoff favoring the Bargainer.

rights an AGI of \$1.00. In our study, strong property rights produced an AGI of \$1.10, which is significantly higher than the AGI of \$0.23 under weak property rights ($p = 0.094$, one-sided t -test).

Table 4. Comparing the Impact of Entitlement and Strength of Property Rights on Payoff Distributions in Our Replication to those HS (1985).

	Strong Property Rights		Weak Property Rights	
	<i>HS Data</i>	<i>Our Data</i>	<i>HS Data</i>	<i>Our Data</i>
AGI	\$4.52	\$1.10	\$1.00	\$0.23

Notes: AGI is the average of the difference between a Controller’s final payoff and what they would have earned from choosing an equal split of the total payoff for that decision outcome. We use 22 observations to calculate AGI for each HS panel, which come from the strong property right sessions (game entitlement plus entitlement priming) and weak property rights sessions (random entitlement plus no entitlement priming) in HS (1985). Results in each of the ‘Our Data’ columns include data from the first two periods in each respective set of two-shot treatment sessions.

To summarize, we successfully replicate several key findings from HS, including the consistent achievement of efficient allocations by subjects, equally high efficiency in one-shot and two-shot bargaining, and that efficiency is invariant to the strength of property rights. We also replicate the result that strong property rights yield a higher AGI. However, we fail to replicate the finding that strong property rights reduce the proportion of sharing allocations, and we do not observe a difference in sharing behavior between one-shot and two-shot bargaining, as HS did.

3 Digital Extension

3.1 Modifications of Procedures

We used the same lab as in the face-to-face sessions, leaving an empty computer station between subjects. The same moderator read instructions aloud for each session, and paper instructions were provided for reference.¹⁸ Instructions included a comprehension quiz checked individually before proceeding. All digital bargaining sessions were conducted within a computer interface programmed in ZTree (Fischbacher,

¹⁸ See Appendices B.1 through B.4 for the instructions used in all digital sessions, which include screenshots of the bargaining interface at all stages.

2007). Our program used previously generated complete-stranger matches for each period for one-shot sessions and every two periods for two-shot sessions.

The digital bargaining process was designed to replicate face-to-face bargaining as closely as possible. After Controllers selected a unilateral decision for implementation in cases of bargaining impasse, Bargainers were informed of this decision. Subjects then used a chat box to bargain. If subjects reached a mutual decision, both parties could indicate this with a button on the chat screen, after which the Bargainer completed a contract and forwarded it to the Controller for final approval. Controllers could reject a contract for any reason. If a Controller refused a contract or did not engage in bargaining, the program implemented the Controller's unilateral decision, and the period ended. If the Controller approved the contract, the program implemented payoffs according to the contract terms, and the period ended.

Assigning Property Rights

We allocated *weak* property rights at the pair level via random number generation. We allocated *strong* property rights by having subject pairs compete in a simple addition task for time, and the winner was told they had *earned the right* to be the Controller for that period. Though this competition task is different than the one employed during face-to-face bargaining, we observe no difference in the frequency of role switching as a result. We chose a programmable task that we thought best replicated the deterministic, competitive properties of the hash mark game described above.

3.2 Results: Digital vs. Face-to-Face

We now test the robustness of bargaining behavior in the digital environment compared to face-to-face. As in Section 2.3, we use Fisher's exact tests to assess statistical differences in the proportions of efficient and sharing allocations both within and across bargaining environments. Table 5 presents the results of the digital extension, based on the first two decisions made by each subject pair in all sessions.¹⁹ We first examine the effect of repeated bargaining on behavior, as in HS (1982).

¹⁹ Power calculations indicate that a sample size of 95 observations per treatment is needed to detect an effect similar to our replication result for sharing allocations between the one-shot and two-shot treatments, while a sample size of

Result 5: Similar to our face-to-face replication, in the digital environment we observe the same efficiency results found in HS (1982). No significant difference in the proportions of *efficient* outcomes is observed between one-shot and two-shot digital bargaining ($p = 0.55$). However, when comparing across bargaining environments, we find significantly lower *efficiency* in *both* one-shot ($p = 0.04$) and two-shot ($p = 0.02$) digital bargaining.

Result 6: Unlike our face-to-face replication but consistent with HS (1982), repeated bargaining in the digital environment induces more *other-regarding behavior* among Controllers. Fisher's exact test shows a significant difference in the proportion of *sharing allocations* between one-shot and two-shot digital bargaining ($p = 0.04$). When comparing across bargaining environments, we observe significantly less *other-regarding behavior* in one-shot digital bargaining ($p = 0.04$), but no significant difference in the two-shot treatments across environments ($p = 0.36$).

Next, we examine the effect of entitlement to property rights on behavior in two-shot bargaining (also presented in Table 5), as studied in HS (1985).

Result 7: Consistent with our face-to-face replication and HS (1985), we find equally high levels of *efficiency* in both strong and weak property rights treatments in the digital environment ($p = 0.39$). However, we observe a significant difference in the proportions of *sharing allocations* between these treatments ($p < 0.01$). Hence, as in HS (1985), the strength of property rights does not affect *efficiency* but reduces the proportion of *sharing allocations*. Comparing across bargaining environments, we find no significant difference in *efficiency* under strong property rights ($p = 1.00$), but significantly lower *efficiency* in the digital environment under weak property rights ($p = 0.02$). In terms of self-regarding behavior, digital bargaining significantly reduces *sharing allocations* under strong property rights ($p < 0.01$) but there is no significant difference under weak property rights ($p = 0.36$).

46 observations per treatment is needed to detect an effect similar to our replication result for sharing allocations between the strong and weak property rights treatments, both to achieve 80% power at a 5% significance level. Consequently, we are underpowered in each case without including more periods or pooling across treatments.

Table 5: Comparing the Impacts of Repeated Bargaining and the Strength of Property Rights on Efficiency and Sharing Allocations in Our Replication to Our Digital Extension.

			Efficient Decisions			Sharing Decisions		
Bargaining Sequence	Property Rights	N _{FtF} N _{Dig}	Face-to-Face	Digital	Fisher's Exact	Face-to-Face	Digital	Fisher's Exact
Repeated Bargaining								
1-Shot	Weak	24 24	15 (.63)	7 (.29)	$p = 0.04$	14 (.58)	6 (.25)	$p = 0.04$
2-Shot	Weak	24 24	19 (.79)	10 (.42)	$p = 0.02$	18 (.75)	14 (.58)	$p = 0.36$
Fisher's Exact	1-Shot vs. 2-Shot		$p = 0.34$	$p = 0.55$		$p = 0.36$	$p = 0.04$	
Strength of Property Rights								
2-Shot Bargaining	Strong	24 24	15 (.63)	14 (.58)	$p = 1.00$	12 (.50)	2 (.08)	$p < 0.01$
2-Shot Bargaining	Weak	24 24	19 (.79)	10 (.42)	$p = 0.02$	18 (.75)	14 (.58)	$p = 0.36$
Fisher's Exact	Strong vs. Weak		$p = 0.34$	$p = 0.39$		$p = 0.14$	$p < 0.01$	

Notes: This table presents the number of Pareto efficient and sharing decisions (proportion of total decisions in parentheses) by session type in both our face-to-face and digital experiments using the first two periods of bargaining in all cells, with p -values for Fisher's exact tests. Our results in the repeated bargaining panel include data from our weak property rights, no entitlement priming sessions across each bargaining environment ($N = 24$ for each one-shot and two-shot cell in each bargaining environment). Our results in the strength of property rights panel include data from our two-shot, strong and weak property rights cells across each bargaining environment ($N = 24$ for each strong and weak property rights cell in each bargaining environment).

Finally, we use the AGI (HS 1985) to assess the effect of entitlement to property rights on self-regarding behavior in the two environments, presented in Table 6.

Result 8: As measured by the AGI, we find that strong property rights induce *more* self-regarding behavior, regardless of whether subjects bargained face-to-face or digitally. However, digital bargaining *amplifies* self-regarding behavior. Strong property rights produced an AGI of \$3.89 in digital sessions (compared to \$1.10 face-to-face, $p < 0.001$), and weak rights an AGI of \$0.93 (compared to \$0.23 face-to-face, $p = 0.17$), with a pooled difference of \$1.73 across environments ($p < 0.001$), using one-sided t -tests for all differences.

Table 6. Comparing the Impact of Entitlement and the Strength of Property Rights on Payoff Distributions in Our Replication to Our Digital Extension.

	Strong Property Rights		Weak Property Rights		Pooled	
	Face-to-Face	Digital	Face-to-Face	Digital	Face-to-Face	Digital
AGI	\$1.10	\$3.89	\$0.23	\$0.93	\$0.67	\$2.41

Notes: AGI is calculated as the average difference between a Controller's final payoff and the payoff they would have received from choosing an equal split of the total payoff for that decision outcome. We use data from the first two periods (N = 24) to compute AGI for the strong property right sessions (game entitlement plus entitlement priming) and weak property rights sessions (random entitlement plus no entitlement priming) across both the face-to-face and digital bargaining environments. Results in the 'Pooled' column were calculated using pooled data from the first two periods in each bargaining environment (N = 48 for each respective cell).

Our digital extension reveals several key differences compared to face-to-face bargaining. While *efficiency* remains equally high across one-shot and two-shot digital treatments (as in our face-to-face replication), digital bargaining significantly reduces *efficiency* in both one-shot and two-shot treatments compared to face-to-face. Unlike in our face-to-face replication, repeated bargaining in the digital environment increases *other-regarding* behavior among Controllers, though digital bargaining only reduces other-regarding behavior in the one-shot (not two-shot) treatment compared to face-to-face. Regarding property rights, *efficiency* remains unaffected by their strength in the digital environment (as in our replication), but digital bargaining only reduces *efficiency* under the weak property rights (not strong) treatment compared to face-to-face.²⁰ Additionally, strong property rights reduce *sharing allocations* in the digital environment (unlike our replication), and digital bargaining reduces *sharing allocations* under the strong property rights (not weak) treatment compared to face-to-face. Lastly, digital bargaining yields a higher AGI in the strong property rights treatment.

²⁰ The latter effect is driven by a sharp rise (drop) in efficiency among digital (face-to-face) bargainers in the second period of the 2-shot, strong property rights treatment, which is not observed in any other treatment (Figures D3 and D4, Appendix D.2).

3.3 Results: Digital vs. Face-to-Face Using Data from All Periods

We now analyze data from all 10 periods of bargaining in each treatment to econometrically assess the robustness of findings on efficiency and sharing allocations in the digital bargaining environment.²¹ First, we examine efficiency, followed by other-regarding behavior.

Efficiency

Table 7 presents the results from Probit and linear probability models to estimate the effect of digital bargaining on the likelihood of achieving a Pareto efficient allocation by a subject pair. Three methods are used to calculate standard errors for the linear probability models, and two for the Probit models.

Result 9: Digital bargaining decreases the likelihood of achieving an efficient outcome by 23.3 percent ($p < .001$). The reduction in efficiency is consistent across all four treatments in the digital environment, with the proportion of efficient decisions ranging from 58 to 75 percent, compared to 88 to 93 percent in face-to-face sessions (Table D1, Appendix D.1).²² The decrease in efficiency remains robust even with increased experience (Figures D1 and D2, Appendix D.2). However, when disaggregated by treatment cell, the effect with increased experience is mostly robust. For example, in the weak property rights, two-shot bargaining treatment, efficiency rates in the digital environment converge to those in face-to-face bargaining by period 5 (Figures D3 and D4, Appendix D.2).

²¹ Table D1 in Appendix D provides the number of Pareto efficient and sharing decisions by session type in both our face-to-face and digital experiments using all periods of bargaining, along with p -values for Fisher's exact tests.

²² Further discussions surrounding these modeling approaches and results are available in Section D.2 of Appendix D.

Table 7. The Effects of Digital Bargaining on Pareto Efficiency, Using Data from All Periods.

	(1)	(2)	(3)	(4)	(5)
	Probit Models		Linear Probability Models		
Digital	-0.233***	-0.233***	-0.233***	-0.233***	-0.233***
	(0.0224)	(0.0382)	(0.0239)	(0.0406)	[-0.3219, -0.1335]
Weak Property Rights	-0.0235	-0.0235	-0.0208	-0.0208	-0.0208
	(0.0233)	(0.0397)	(0.0239)	(0.0406)	[-0.1129, 0.083]
2-Shot Bargaining	0.0461**	0.0461	0.05**	0.05	0.05
	(0.0233)	(0.0388)	(0.0239)	(0.0406)	[-0.0407, 0.1531]
Period FEs	Yes	Yes	Yes	Yes	Yes
Errors	Robust	Robust, Clustered	Robust	Robust, Clustered	Wild Bootstrap
N	960	960	960	960	960

Notes: This table reports the results from a series of regressions using Probit models (columns 1 and 2, showing average marginal effects) and linear probability models (columns 3 through 5). In each regression, the dependent variable is an indicator of whether a bargaining outcome was Pareto efficient, projected onto a set of indicator variables that describe the bargaining environment. “Digital” indicates whether the bargaining decision occurred digitally. “Weak Property Rights” indicates whether property rights were randomly assigned to the Controller without entitlement priming. “2-Shot Bargaining” indicates whether the decision occurred in the two-shot bargaining treatment. The estimating dataset includes decisions from all 10 periods in all treatments, and we include period fixed effects in all specifications. Columns (1) and (3) report robust standard errors, (2) and (4) report robust errors clustered at the session level, and (5) uses Wild bootstrapping with 2000 repetitions (confidence intervals provided in brackets). Statistical significance is denoted as follows: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Payoff Distributions

Table 8 presents the results from Probit and OLS models, again using various methods to calculate standard errors, to estimate the effect of digital bargaining on the likelihood of observing an equal split, and the levels of greed and sacrifice exhibited by Controllers.

Result 10: Controllers are more self-regarding and individually rational when bargaining digitally. We find that digital bargaining reduces the likelihood of a bargaining pair reaching an equal split by 34.7% ($p < 0.01$, columns 1 and 2), nearly quadruples greed ($p < 0.001$, columns 3 and 4, where the AGI increases to 2.96 from a baseline of 0.73 in face-to-face bargaining, Table D2, Appendix D.2), and reduces the amount

of money Controllers are willing to sacrifice relative to their unilateral maximum ($p < 0.001$, columns 5 and 6). As measured by the AGI, the increased self-regarding behavior observed in digital bargaining is also robust to experience (Figure D5, Appendix D.2) and holds in all treatments except the weak property rights, two-shot treatment, where we observe no significant difference in AGI between the digital and face-to-face environments across periods (Figure D6, Appendix D.2). This suggests that Controllers are more likely to behave in an individually rational way and are most self-regarding in the digital environment when property rights are clearly defined and or repeated interaction is not a concern. Further discussions regarding this behavior and experiential learning in bargaining are available in Appendix D.

Table 8: The Effects of Digital Bargaining on Allocations, Using Data from All Periods.

	(1)	(2)	(3)	(4)	(5)	(6)
	Equal Split		Greed Index		Sacrifice	
Digital	-0.347*** (0.0202)	-0.347*** (0.0501)	2.234*** (0.3698)	2.234*** [1.4126, 3.0855]	-0.17*** (0.0301)	-0.17*** [-0.2402, -0.1025]
Weak Property Rights	0.338*** (0.0205)	0.338*** (0.0511)	-1.462*** (0.3698)	-1.462*** [-2.3727, -0.5839]	0.127*** (0.0301)	0.127*** [0.056, 0.2011]
2-Shot Bargaining	0.186*** (0.0245)	0.186*** (0.0606)	-0.603 (0.3698)	-0.603 [-1.5164, 0.2899]	0.047 (0.0301)	0.047 [-0.0266, 0.1214]
Period FEs	Yes	Yes	Yes	Yes	Yes	Yes
Errors	Robust	Robust, Clustered	Robust, Clustered	Wild Bootstrap	Robust, Clustered	Wild Bootstrap
N	960	960	960	960	960	960

Notes: Regression results from Probit models are in columns (1) and (2) and pooled OLS models are in columns (3) through (6). Column titles correspond to the dependent variable. The “Equal Split” outcome indicates whether an equal split was observed in each bargaining decision. The “Greed Index” outcome reflects the difference between a Controller’s final payoff and what they would have earned from choosing an equal split of the total payoff. The “Sacrifice” outcome is based on a calculation of the Controller’s sacrifice rate for a bargaining decision. Let U be the unilateral maximum amount available to a Controller, and B be the payoff to the Controller conditional on accepting a Bargainer’s proposal. Then, “Sacrifice” is defined as $S = \frac{U-B}{U}$, representing the percentage of Controller earnings from unilateral maximization that they would sacrifice by accepting the proposal. Each of these outcomes are projected onto a set of indicator variables that describe the bargaining environment. “Digital” indicates whether the bargaining decision occurred digitally. “Weak Property Rights” indicates whether property rights were randomly assigned to the Controller without entitlement priming. “2-Shot Bargaining” indicates whether the decision occurred in the two-shot bargaining treatment. The estimating dataset includes decisions from all 10 periods in all treatments, and we include

period fixed effects in all specifications. As in Table 7, we report results for specifications using several types of standard errors. Column (1) reports robust standard errors, columns (3) and (5) report robust errors clustered at the session level, and (4) and (6) use Wild bootstrapping with 2000 repetitions (confidence intervals provided in brackets). Statistical significance is denoted as follows: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5 Discussion and Conclusion

This paper replicates and extends the seminal Coasian bargaining experiments introduced in HS (1982; 1985), using a 2x2x2, between-subjects design. Our design varies the method of assigning property rights, whether subjects engaged in repeated bargaining, and whether bargaining occurred face-to-face or digitally. Like HS, we use entitlement priming (with strong property rights) or neutral language (with weak property rights) to further distinguish the strength of property rights in our treatments.

Our results from face-to-face bargaining are mostly consistent with those in HS (1982, 1985). Efficiency remains equivalently high across all four treatments, and we observe a higher AGI when bargaining occurs with strong property rights compared to weak property rights. However, we do not replicate HS’s finding that one-shot bargaining leads to more self-regarding behavior than two-shot bargaining.²³ When testing the robustness of these results in a digital bargaining environment, we observe a significant decline in the ability of subjects to achieve gains from trade and less equal surplus distribution in digital bargaining compared to face-to-face. These results suggest that Coase’s theorem may require additional behavioral considerations, particularly when negotiations occur under the veil of anonymity.²⁴

Several procedural differences between the face-to-face and digital environments may have contributed to the observed variation in behavior. In the digital environment, Controllers preselected a disagreement outcome, the announcement of which could potentially be interpreted by counterparts as a more salient or antagonistic threat compared to the face-to-face setting where preselection was possible but not required. The tasks used to determine strong property rights also varied slightly: the addition task used

²³ We cannot rule out that this is driven by a tit-for-tat strategy.

²⁴ This notion aligns with the idea that what HS and Harrison and McKee (1985) identified as other-regarding behavior may actually be driven by self-regarding motivations, specifically the desire to avoid interpersonal conflict. Such preferences could be shaped by psychological and confrontation costs, including feelings of awkwardness, embarrassment, or guilt (Jindal and Newberry 2018; Gago 2024), which are more pronounced in face-to-face bargaining settings.

in digital sessions may have required more effort, as opposed to the potentially more luck-based hash mark game in face-to-face sessions, which may have strengthened perceived entitlement in digital Controllers. Additionally, while both environments followed the same subject payment structures, minor variations in the randomly chosen periods for payments across one-shot and two-shot treatments could have influenced behavior. Finally, face-to-face subjects negotiated in the same room, where reputational concerns or social pressure from future interactions may have encouraged more equal bargaining outcomes, compared to the more anonymous digital environment.

We believe these findings have practical implications. Face-to-face interactions appear more effective for resolving resource allocation disputes, such as those encountered in legal arbitration or contract negotiation (e.g., Fenn and Rickman 1999). In contrast, digital bargaining makes property rights holders more rigid, which places increased pressure on counterparts to make precise offers or accept reduced payoffs. Moreover, non-property rights holders do not sufficiently adjust their bargaining approach or expectations about allocations, which suggests that reducing social distance among team members could improve task allocation and overall productivity.

Although our study sheds light on how digital environments alter bargaining behavior, future research is needed to uncover the full set of mechanisms underlying our results. First, it is important to investigate whether the lower efficiency observed in digital bargaining stems from a higher frequency of disagreements, or an inability to reach efficient agreements. Second, future studies should analyze the share of unilateral decisions and the proportion of those decisions that maximize Controller payoffs to reveal if they are more generous in face-to-face settings even when making unilateral decisions. Finally, examining differences in earnings between Controllers and Bargainers, particularly in non-unilateral decisions, could indicate whether Controllers aim for equitable payoffs in both environments but are less successful online.

7 References

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